Parametric & Sequences Test

Cram sheet

Wed, no calculator: 5 parametric curve questions + 1 sequence question. **Fri, calculator:** 7 sequences & series + 1 projectile. 43 points.

Memorize: not given on the test

(1) $\sum ca_k = c\sum a_k$

(2), (3) $\sum(a_k \pm b_k) = \sum a_k \pm \sum b_k$

(4) $\sum_{k=j+1}^{n} a_k = \sum_{k=1}^{n} a_k - \sum_{k=1}^{j} a_k$ (j = start -1)

(5) $\sum_{k=1}^{n} 1 = n$ (6) $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$

(7) $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$

(8) $\sum_{k=1}^{n} k^3 = \left[\frac{n(n+1)}{2}\right]^2$

$0! = 1,\ 1! = 1$ • $\sin^2 t + \cos^2 t = 1,\ 1 + \tan^2 t = \sec^2 t,\ 1 + \cot^2 t = \csc^2 t$ • Given: $x = (v\cos\theta)t,\ y = (v\sin\theta)t - 16t^2 + h$

Find a sum (Friday)

1. Lower index not 1: **fix it with (4) first**. Start at $k = 6$ means $j = 5$.
2. Split with (2), (3); constants out with (1).
3. Write each formula from memory with n plugged in. **This line is required before the calculator.** The Σ button is blocked in test mode.

Ex: $\sum_{k=6}^{20} 4k^2 = 4\left[\frac{20(21)(41)}{6} - \frac{5(6)(11)}{6}\right] = 11260$ $\sum_{k=1}^{10}(k^3+1) = \left[\frac{10(11)}{2}\right]^2 + 10 = 3035$

Binomial Theorem: Pascal's triangle (Friday)

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
1 6 15 20 15 6 1
1 7 21 35 35 21 7 1
1 8 28 56 70 56 28 8 1

- Outsides 1; inside = sum of the two above. Row n for power n (up to about 7 or 8).
- Blanks first: first term's powers **down** $n \to 0$, second term's **up** $0 \to n$. Fill with row n, simplify.
- The sign goes with the second term; odd powers of a negative are negative.
- $(2x)^4 = 16x^4$, not $2x^4$.

Ex: $(2x-3)^4 = 16x^4 - 96x^3 + 216x^2 - 216x + 81$

Sequences

- **Terms:** plug in $n = 1, 2, 3, \ldots$ Answer is a list with commas, reduced, no ellipsis.
- ***n*th term:** write 1 2 3 4 above; top and bottom separately; first term positive $(-1)^{n+1}$, negative $(-1)^n$; odds $2n-1$, squares n^2, powers b^{n-1}. Check $n = 1$. Ex: $\frac{1}{3}, -\frac{2}{9}, \frac{3}{27}, \ldots \to a_n = (-1)^{n+1}\frac{n}{3^n}$
- **Recursive:** first term given; the formula starts where it says ($n \geq 2$); a_{n-1} is the previous term.
- **Write out a sum:** first few terms, ellipsis, last term; don't add. **Express as Σ:** lower index, upper index, general term, same letter. Check $k =$ lower index gives the first term.

Parametric curves (Wednesday)

1. **t-chart first.** Brackets: plug in both ends, solid dots. Parentheses: values inside.
2. Solve for t where it gives **one answer** (no $\pm$); substitute into the other.
3. Domain: smallest and largest x in the chart, "for x in $[a,b]$". Watch turning points ($x=t^2\geq 0$, $x=\sqrt{t}\geq 0$).
4. Any form is fine once t is gone; no multiplying out. **SIMPLIFIED** means y alone. No domain written means all real x.
5. Direction arrows **only when the question asks**: lowest t to highest. End arrowheads only when the curve keeps going. Asymptotes optional, dashed if drawn.

Trig curve	**Final form**
full circle / ellipse	$\frac{x^2}{9}+\frac{y^2}{25}=1$, no domain
one hyperbola branch (restricted t)	solve for y, keep that branch's sign ($+\sqrt{}$ top, $-\sqrt{}$ bottom): $y=2\sqrt{\frac{x^2}{9}+1}$; never "$y\geq$"; any equivalent form
both branches (no restriction)	$y=\pm 3\sqrt{\frac{x^2}{4}-1}$: $\pm$ **!!!**
reciprocal $x=t+2,\ y=\frac{1}{t+2}$	$y=\frac{1}{x}$; "for all $x\neq 0$" is good practice (it comes with the equation)

Two parametric representations

$y=(x-2)^2+1$: $\quad x=t,\ y=(t-2)^2+1\quad$ and $\quad x=t+2,\ y=t^2+1,\quad$ for t in $(-\infty,\infty)$ (or the restricted interval). Not in vertex form? Complete the square first.

Projectile (Friday)

1. Equations with the numbers and the degree symbol; from the ground $h=0$.
2. Degree mode, Graph, Parametric, $x_1(t)$, $y_1(t)$ exactly; write the setup. Path doesn't come down? Raise $t$max.
3. Max height: top, read y. Time: $y\approx 0$, read t. Distance: x at that point. Trace step 0.01, then 0.001.
4. Thousandths, units in feet; "$y\approx$", "$t\approx$", "$x\approx$" are full answers: $y\approx 26.258$ ft, $t\approx 2.512$ s, $x\approx 126.707$ ft. Other way: quadratic formula on $0=-16t^2+(v\sin\theta)t+h$.

Where points get lost

- Dropping $\pm$ on an unrestricted sec/tan hyperbola.
- Arithmetic slips while simplifying: check one t-chart point in the final equation.
- Using a sum formula before fixing a lower index that isn't 1; skipping the substituted-formula line.
- $(2x)^n$ written as $2x^n$; the minus sign left off the second term.
- Wrong sign exponent on alternating sequences: check a_1.